DAY 1

1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

```
+-----------------------------+
|          MOTHERBOARD        |
|                             |
|   +------+                  |
|   | CPU  |                  |
|   +------+                  |
|                             |
|  | RAM |   | RAM |          |
|                             |
+-----------------------------+
         |
         | GPU (Graphics Card)
    ====================

 +-----------+    +-----------+
 |    HDD    |    |    SSD    |
 +-----------+    +-----------+
```

Motherboard – Main circuit board that connects all components.

CPU (Central Processing Unit) – Performs calculations and processing.

RAM (Random Access Memory) – Temporary memory used while programs run.

GPU (Graphics Processing Unit) – Handles graphics and video output.

HDD (Hard Disk Drive) – Traditional storage device.

SSD (Solid State Drive) – Faster storage device with no moving parts.

**2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).**

HDD (Hard Disk Drive)	SSD (Solid State Drive)
Slower read/write speeds due to moving mechanical parts	Much faster read/write speeds because it uses flash memory
Less durable; can be damaged by shocks or drops	More durable and shock-resistant since it has no moving parts
Takes longer to start the operating system	Boots the operating system much faster
Produces some noise while operating	Silent operation
Uses more power	Uses less power, improving battery life
Usually available in larger capacities at lower cost	Typically more expensive per GB, though capacities are increasing
Desktop PCs, file servers, large media storage, backups	Laptops, gaming PCs, ultrabooks, high-performance computers
Cheaper per gigabyte	More expensive per gigabyte

3.Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU (Central Processing Unit):

The CPU is the "brain" of the computer. It processes game logic, controls player actions, manages physics, and coordinates communication between different hardware components to keep the game running smoothly.

RAM (Random Access Memory):

RAM temporarily stores the game data that is currently being used, such as maps, character information, and textures. Having enough RAM allows games like PUBG or Free Fire to load faster and run without lag caused by constant loading from storage.

GPU (Graphics Processing Unit):

The GPU is responsible for rendering the game's graphics, including characters, environments, lighting, and visual effects. A powerful GPU provides smoother gameplay, higher frame rates, and better image quality, making the gaming experience more enjoyable.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

PC Hardware Upgrades for Running the Latest FIFA Smoothly

1. **Graphics Card (GPU):**
   I would upgrade the GPU because it renders the game's graphics. A better GPU provides smoother gameplay, higher frame rates, and improved visual quality.

2. **Processor (CPU):**
   I would upgrade the CPU because it handles game calculations, player movements, and AI. A faster processor reduces lag and improves overall game performance.

3. **RAM (Memory):**
   I would upgrade the RAM to at least **16 GB** so the game can load and run smoothly. More RAM also helps when running other programs in the background.

4. **Storage (SSD):**
   I would replace an HDD with an **SSD** because it loads the game much faster and improves the computer's overall speed.

5. **Power Supply Unit (PSU):**
   I would upgrade the PSU if the new components require more power. A reliable PSU ensures the PC runs safely and stably.

6. **Cooling System:**
   I would improve the cooling system to keep the CPU and GPU from overheating during long gaming sessions. Better cooling helps maintain good performance and extends the life of the components.